CareConnect EHR System — Project Demonstration Script

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GitHub Repository: https://github.com/scrollingreel/P_003-CareConnect-Patient-EHR-HCL-TECH-PROJECT

Step-by-step evaluator script for demonstrating all seven (7) mandatory HCL Epics in the live executing CareConnect application.

Evaluator Demo Personas & Credentials

- **Doctor Persona:** `doctor@apexcare.org` / `DoctorPass2026!` (Dr. Rajesh Verma - Cardiology)
- **Patient Persona:** `ananya.sharma@example.com` / `SecurePass2026!` (Ananya Sharma - ID: pat-101)
- **Admin Persona:** `admin@apexcare.org` / `SecurePass2026!` (Amanda Ross - Hospital Operations)

10-Step Comprehensive Demonstration Walkthrough

1. **Step 1: Portal Landing & Authentication (Epic 6 & Security)**
   Navigate to `http://localhost:3000/login`. Demonstrate the quick persona switcher buttons (Doctor, Patient, Admin). Click "Doctor" to auto-fill credentials and log in.
2. **Step 2: Doctor Workstation & Patient Directory (Epic 1)**
   From the Doctor Dashboard, review the schedule and active queues. Click "Patients" in the sidebar to view the directory. Select **Ananya Sharma (pat-101)** to open the electronic health chart.
3. **Step 3: Diagnostic Results & Critical Panic Alerts (Epic 5)**
   In the patient chart, click the **Diagnostic Reports** tab. Highlight the quantitative findings and the red critical panic alert banner for Serum Potassium (6.4 mmol/L). Click the **"HL7 / FHIR Payload"** button to inspect live serialized HL7 v2.5.1 ORU^R01 segments and FHIR R4 JSON.
4. **Step 4: CPOE Multi-Category Order Entry (Epic 3)**
   Click the **Diagnostic & Clinical Orders** tab. Review existing orders and status progression. Click **"Place New Clinical Order"**. Select Order Type: *Laboratory Test*, choose *Cardiac Troponin I*, set Urgency to *STAT*, and submit. Observe the new order appearing in the active tracking table.
5. **Step 5: Medication Prescribing & DDI Contraindication Screening (Epic 4)**
   Click the **Prescriptions & MAR** tab. Note that the patient is actively taking Lisinopril. Propose prescribing *Spironolactone*. Observe the real-time red warning banner flagging high-risk hyperkalemia contraindication.
6. **Step 6: Medication Administration Record (MAR) Logging (Epic 4)**
   Scroll to the MAR log. Click **"Record Administration"**. Record a dose of Atorvastatin as *GIVEN* by Nurse Meera. Verify the verified timestamp and clinician signature.

7. **Step 7: Clinical Documentation & Tamper-Resistant Locking (Epic 2)**
   Click the **Clinical Notes** tab. Select template *SOAP Note*. Review the distinct **Assessment** and **Treatment Plan** fields. Show a signed note with the green "*Finalized & Signed*" lock badge. Explain that the backend enforces HTTP 409 Conflict if modifications are attempted on finalized records.

8. **Step 8: Patient Portal & Released Health Records (Epic 7)**
   In the top navigation bar, click **Patient (Ananya)** to switch to the patient persona. View the Patient Dashboard showing upcoming visits and released test summaries. Click **Health Records** to demonstrate read-only access.

9. **Step 9: Secure Patient-Provider Messaging (Epic 7)**
   On the Patient Dashboard, locate the **Secure Messaging** section. Send a message to Dr. Rajesh Verma regarding follow-up lab timing. Verify the message is saved to the persistent thread.

10. **Step 10: Admin Analytics & Immutable Audit Trail (Epic 6)**
    Click **Admin (Operations)** in the top navbar. Navigate to `/admin/analytics` to review hospital throughput KPIs and inspect the chronological `audit_logs` recording every clinical action performed during the demo.